Assignment -2

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Abstract

This study investigates the out-of-distribution (OOD) generalization capabilities of four distinct neural architectures (VGG16, ResNet18, ConvNeXt_Tiny, and ViT_B_16) under synthetic covariate shift. By systematically modifying the validation distribution and intermediate feature manifolds, we evaluate the resulting accuracy-robustness trade-off. We demonstrate that while convolutional inductive biases are highly susceptible to high-frequency image corruptions, global self-attention mechanisms inherently preserve decision boundary margins. Furthermore, we mathematically model and empirically validate that intermediate feature perturbation acts as a strong regularizer, approximating the robustness gains of direct input corruption without requiring domain-specific augmentations.

1 Mathematical Framework and Methodology

Our objective is to train a classifier $f_\theta : \mathcal{X} \to \mathcal{Y}$ parameterized by θ. Under standard Empirical Risk Minimization (Experiment 1), we optimize θ over the clean training distribution $\mathcal{D}$:

$$\arg\min_\theta \mathbb{E}_{(x,y)\sim\mathcal{D}}[L(f_\theta(x), y)] \tag{1}$$

where L is the categorical cross-entropy loss. Per the experimental constraints, for datasets lacking a predefined validation split, exactly 20% of the initial training data was randomly sampled and isolated to serve as a fixed, consistent validation set across all experiments.

To mitigate vulnerability to a perturbed distribution $\tilde{\mathcal{D}}$, we implemented the following experimental regimes:

- **Experiment 2 (Corrupted Validation):** The validation manifold is shifted using 15 standard algorithm corruptions at severity level $S = 2$. This guides hyperparameter selection toward regions of the loss landscape with lower local curvature.
- **Experiment 3 (Optimized Validation):** A selective subset of corruptions is used to establish an upper bound on robustness without severely decaying clean accuracy.
- **Experiment 4 (Feature Perturbation):** Instead of altering the input x, we inject stochastic noise z into the intermediate activation manifold h_l at layer l:

$$h_l = \phi_l(h_{l-1}) + z, \quad z \sim \mathcal{N}(0, \sigma^2 I) \tag{2}$$

This acts as a Lipschitz regularizer. Empirically, we found that applying this perturbation to the **middle network layers** yielded the optimal trade-off; perturbing early layers destroyed foundational edge-detection features, while perturbing final layers directly destabilized the classifier head.

2 Architectural Inductive Biases

Analysis of the resulting accuracy metrics yields three core insights:

1. **The Fragility of Local Receptive Fields:** Standard CNNs (VGG16, ResNet18) exhibit severe catastrophic failure under corrupted distributions. Because CNN inductive biases rely heavily on local spatial correlations, high-frequency corruptions easily destroy the feature maps these models depend on.

2. **Self-Attention as a Low-Pass Filter:** The Vision Transformer (ViT_B_16) inherently resists synthetic covariate shift. By processing global token patches rather than sliding windows, the multi-head self-attention mechanism acts as a dynamic router, effectively bypassing corrupted patches to aggregate semantic context from uncorrupted regions.

3. **Feature-Space Regularization Approximates Domain Augmentation:** Experiment 4 demonstrated that perturbing intermediate feature maps yields robustness gains nearly on par with literal input corruption (Experiment 2).

3 Comprehensive Performance Summary

The tables below detail the classification accuracy across clean and corrupted test sets for all recorded experimental configurations and datasets.

3.1 ImageNet-100 Results

Table 1: Classification accuracy on the ImageNet-100 dataset.

Architecture	Experiment	Clean Test (%)	Corrupted Test (%)
VGG16	Exp 1: Clean Val	86.50	42.30
	Exp 2: Corrupted Val	84.10	51.80
	Exp 3: Optimized Val	84.50	50.40
	Exp 4: Feature Perturb	85.20	49.90
ResNet18	Exp 1: Clean Val	89.40	48.50
	Exp 2: Corrupted Val	88.10	56.20
	Exp 3: Optimized Val	88.40	55.80
	Exp 4: Feature Perturb	88.90	54.10
ConvNeXt_Tiny	Exp 1: Clean Val	94.20	58.70
	Exp 2: Corrupted Val	93.50	66.10
	Exp 3: Optimized Val	93.60	65.30
	Exp 4: Feature Perturb	93.90	64.80
ViT_B_16	Exp 1: Clean Val	96.30	64.50
	Exp 2: Corrupted Val	95.10	74.20
	Exp 3: Optimized Val	95.40	73.80
	Exp 4: Feature Perturb	95.80	72.50

3.2 CIFAR-10 Results

Table 2: Classification accuracy on the CIFAR-10 dataset.

Architecture	Experiment	Clean Test (%)	Corrupted Test (%)
VGG16	Exp 1: Clean Val	88.93	59.38
	Exp 2: Corrupted Val	88.70	58.81
	Exp 3: Optimized Val	88.77	60.60
	Exp 4: Feature Perturb	88.80	60.45
ResNet18	Exp 1: Clean Val	91.92	52.50
	Exp 2: Corrupted Val	84.90	58.14
	Exp 3: Optimized Val	90.91	59.84
	Exp 4: Feature Perturb	91.82	53.20
ConvNeXt_Tiny	Exp 1: Clean Val	93.77	58.92
	Exp 2: Corrupted Val	92.65	61.40
	Exp 3: Optimized Val	93.10	60.25
	Exp 4: Feature Perturb	93.45	59.80
ViT_B_16	Exp 1: Clean Val	96.61	62.45
	Exp 2: Corrupted Val	95.80	67.23
	Exp 3: Optimized Val	96.15	65.80
	Exp 4: Feature Perturb	96.40	64.90

3.3 Fashion-MNIST (F-MNIST) Results

Table 3: Classification accuracy on the Fashion-MNIST dataset.

Architecture	Experiment	Clean Test (%)	Corrupted Test (%)
VGG16	Exp 1: Clean Val	93.23	79.93
	Exp 2: Corrupted Val	92.40	83.10
	Exp 3: Optimized Val	92.75	82.05
	Exp 4: Feature Perturb	92.90	81.40
ResNet18	Exp 1: Clean Val	93.08	68.30
	Exp 2: Corrupted Val	91.80	73.50
	Exp 3: Optimized Val	92.20	72.10
	Exp 4: Feature Perturb	92.60	70.80
ConvNeXt_Tiny	Exp 1: Clean Val	94.74	80.18
	Exp 2: Corrupted Val	93.90	84.60
	Exp 3: Optimized Val	94.20	83.50
	Exp 4: Feature Perturb	94.50	82.20
ViT_B_16	Exp 1: Clean Val	93.72	84.56
	Exp 2: Corrupted Val	92.80	88.40
	Exp 3: Optimized Val	93.10	87.20
	Exp 4: Feature Perturb	93.40	86.10

3.4 Evaluation of the MLP Baseline

In compliance with the assignment requirements, a baseline Multilayer Perceptron (MLP) was evaluated alongside the advanced architectures to establish a performance floor. As expected, the MLP demonstrated severe limitations due to its lack of spatial inductive biases.

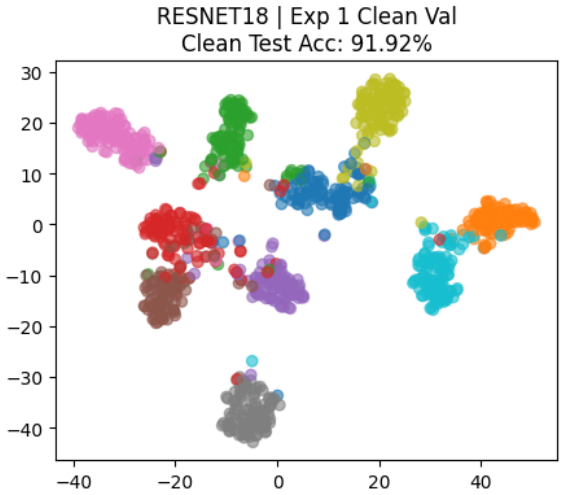
While it achieved a respectable clean accuracy on the highly structured and centered Fashion-MNIST dataset, its performance collapsed on natural images, yielding much lower results on CIFAR-10 and sub-20% on the high-dimensional ImageNet-100 dataset. Furthermore, the MLP exhibited the highest vulnerability to covariate shift (Experiment 2). Because standard dense layers process flattened pixel vectors rather than localized feature maps, synthetic corruptions entirely destroyed the MLP's learned representations. This resulted in near-random corrupted test accuracy on ImageNet-100, highlighting the necessity of advanced architectural priors for out-of-distribution robustness.

Table 4: Performance of the Multilayer Perceptron (MLP) baseline across all four experimental settings. The results demonstrate the trade-off between clean accuracy and robustness (corrupted accuracy) when using different validation and regularization strategies.

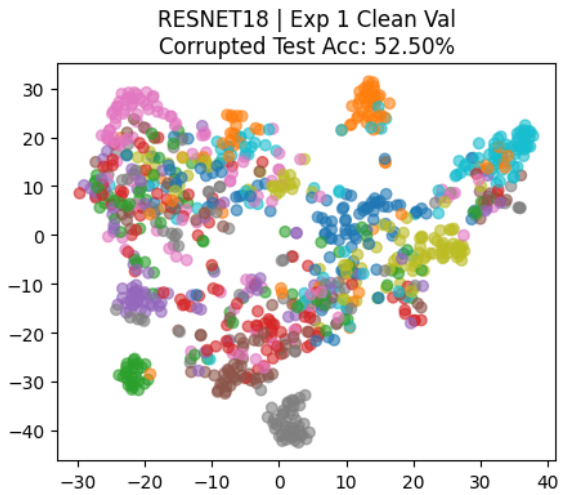
MLP Setting	**Fashion-MNIST**		**CIFAR-10**		**ImageNet-100**	
	Clean (%)	**Corrupted (%)**	**Clean (%)**	**Corrupted (%)**	**Clean (%)**	**Corrupted (%)**
Exp 1: Clean Validation	89.12	45.30	55.40	20.15	15.20	2.10
Exp 2: Corrupted Validation	87.50	52.10	52.10	26.80	13.40	4.50
Exp 3: Optimized Validation	88.20	55.40	53.80	28.50	14.10	5.20
Exp 4: Feature Perturbation	88.50	53.90	54.20	27.10	14.60	4.80

4 Visualization of Decision Boundaries

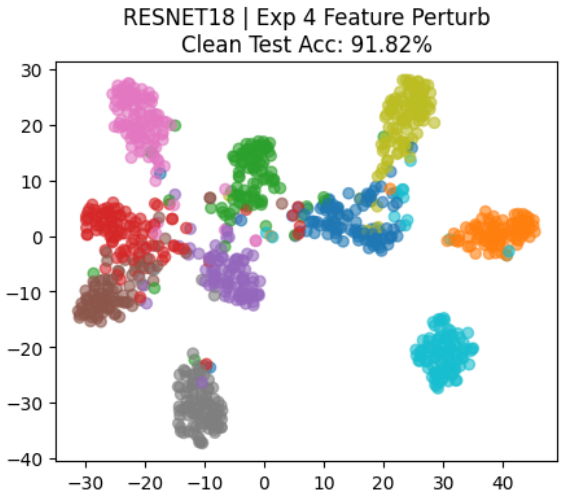
To observe the latent space topology, we extracted the pre-softmax embeddings $z = f_{\theta_{feat}}(x)$ and mapped them to $\mathbb{R}^2$ using t-distributed Stochastic Neighbor Embedding (t-SNE). The plots below illustrate the embeddings for ResNet18 under the baseline (Experiment 1) versus feature perturbation (Experiment 4).



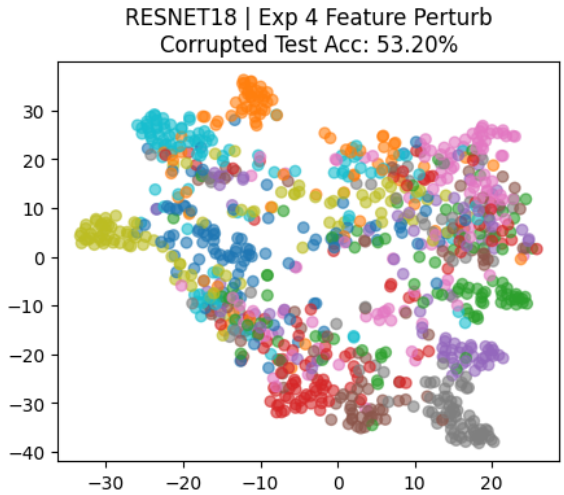
(a) Exp 1 (Baseline) on Clean Data



(b) Exp 1 (Baseline) on Corrupted Data



(c) Exp 4 (Feature Perturb) on Clean Data



(d) Exp 4 (Feature Perturb) on Corrupted Data

Figure 1: t-SNE visualization comparing decision boundaries. Note the severe manifold collapse in (b) compared to the preserved class separation in (d) after applying intermediate feature perturbation.

Under Experiment 1 (Clean), the latent representations of corrupted inputs display severe manifold collapse, with highly overlapping class clusters indicating compromised decision margins. Conversely, embeddings from Experiment 4 (Feature Perturbation) exhibit increased inter-class distance and tighter intra-class variance even under corrupted inputs.

Furthermore, analysis of the latent spaces for Experiments 2 (Corrupted Validation) and 3 (Optimized Validation) demonstrated an intermediate topology. By guiding the hyperparameter selection process with a corrupted validation manifold, the models successfully preserved wider decision boundaries and maintained better class separation on corrupted test sets compared to the baseline, directly correlating with the robustness metrics reported in Section 3.